Microeconomics MID

Show equations, substitutions, intermediate reasoning, and a final labeled answer.

1. [8 pts] Market demand is $Qd=55–3P$ and supply is $Qs=3+P$. Find equilibrium (P^*, Q^*) and show both equations are satisfied.

2. [7 pts] At price $19, quantity demanded is 60; a 10% price increase reduces quantity to 54. Compute arc elasticity and classify demand.

3. [7 pts] Draw or describe the effect of a binding per-unit tax: identify the buyer price, seller price, tax wedge, quantity, and deadweight loss direction.

4. [6 pts] A negative production externality is present. Compare the private and social marginal-cost curves and name one policy that can internalize the cost.

5. [5 pts] Explain why a price ceiling below equilibrium creates a shortage, and identify one non-price allocation mechanism that may follow.

6. [8 pts] Market demand is $Qd=44–4P$ and supply is $Qs=4+P$. Find equilibrium (P^*, Q^*) and show both equations are satisfied.

7. [7 pts] At price $16, quantity demanded is 50; a 10% price increase reduces quantity to 45. Compute arc elasticity and classify demand.

8. [7 pts] Draw or describe the effect of a binding per-unit tax: identify the buyer price, seller price, tax wedge, quantity, and deadweight loss direction.